

Concept Review

Weather Sensors

Why Use Weather Sensors?

Weather sensing is essential through numerous engineering and scientific disciplines. Weather sensors can be found in smartphones for altitude tracking, HVAC systems for climate control, agricultural operations for crop management, and meteorological stations for forecasting. Knowledge of sensor selection, interfacing, and data interpretation provides valuable skills applicable throughout engineering careers, particularly as smart systems and IoT applications continue to expand.

Introduction

Weather sensors measure atmospheric conditions, specifically climate-related parameters that are relevant for weather monitoring or forecasting, including temperature, barometric pressure, humidity, and precipitation. They fall within the broader category of sensors that measure a wide range of environmental factors, such as water quality, air quality, soil pH, and radiation. An environmental sensor is any device that monitors parameters in its surroundings and then transmits this information to a connected system for analysis, tracking, or control.

These sensors function by continuously or periodically monitoring atmospheric conditions and providing data that can be used to make adjustments, track environmental changes, or trigger alerts in automated systems. The ability to accurately measure these parameters is fundamental to applications ranging from personal devices and weather stations to industrial climate control systems.

Types of Weather Sensors

Temperature Sensors

Temperature sensors measure the ambient temperature at a specific location, providing critical data for weather monitoring and forecasting. They can also be used as reference to calibrate other sensors. Three main technologies are commonly used:

Thermistors: Temperature-sensitive resistors that change resistance with temperature. Highly accurate but limited to a narrow temperature range.

Thermocouples: Voltage-generating junctions of dissimilar metals. Wide temperature range, rugged, but require cold junction compensation.

Integrated Circuit (IC) Temperature Sensors: Transistor-based transducers that output voltage or digital signals proportional to temperature. These are very small, accurate, and inexpensive to manufacture. IC temperature sensors are commonly found in consumer electronics like cars and computers due to their balance of cost, size, and precision.

Humidity Sensors

Humidity sensors, also called hygrometers, measure the amount of moisture present in the air, which is a critical parameter for understanding weather conditions and maintaining controlled environments. Humidity measurements are essential for applications ranging from climate control in buildings to assessing plant health in agriculture. These sensors work by detecting how moisture affects the electrical properties of certain materials. The main technologies used are:

Capacitive Sensors: Measure humidity by detecting changes in electrical capacitance as moisture levels vary. These are the most common types due to their good accuracy and relatively fast response.

Resistive Sensors: Measure changes in electrical resistance caused by moisture absorption in a hygroscopic material. Generally, less expensive but may have slower response times.

Barometric Pressure Sensors

Barometric pressure sensors (barometers) measure atmospheric pressure, which is crucial for predicting short-term weather changes and identifying long-term climate trends. Falling pressure typically indicates approaching storms, while rising pressure suggests improving conditions. Beyond weather prediction, barometric pressure sensors enable altitude estimation, indoor navigation, and altitude stabilization in drones and aircraft. Modern electronic pressure sensors employ several technologies:

Piezoresistive: Uses materials whose electrical resistance changes under mechanical stress from atmospheric pressure.

Capacitive: Measures of pressure-induced changes in capacitance between two plates.

Resonant: Detects pressure by measuring changes in the resonant frequency of a vibrating element.

Precipitation Sensors

Precipitation sensors measure the amount and intensity of rainfall, snowfall, or other forms of precipitation. These sensors are essential for hydrological monitoring, flood prediction, and agricultural water management. Common types include:

Optical Rain Sensors: Use infrared beams to detect precipitation by observing how water droplets scatter the light. They can measure both the presence and intensity of rainfall in real-time.

Pressure Altitude Calculation

An important application of barometric pressure sensors is altitude determination. Given the atmospheric pressure at a location, the altitude can be calculated using the International Standard Atmosphere (ISA) model.

The equation to convert atmospheric pressure in millibars (mb) to pressure altitude in feet (ft) is:

$$h_{ft} = 145366.45 \left[1 - \left(\frac{P_{mb}}{P_0} \right)^{0.190284} \right]$$

Where h_{ft} is the pressure altitude in feet, P_0 is the standard sea level pressure (1013.25 mb), and P_{mb} is the measured atmospheric pressure in millibars.

Important Note: Weather conditions significantly affect pressure and altitude calculations. For accurate altitude determination, the actual sea level pressure for the current location and day must be known. Using ISA standard defaults (101,325 Pa and 288 K at sea level) without local calibration can result in altitude errors exceeding 10 meters. Sources: <https://www.weather.gov/media/epz/wxcalc/pressureAltitude.pdf>, <https://www.mide.com/air-pressure-at-altitude-calculator>

BME280 Environmental Sensor

The Mechatronics Sensors Trainer contains the Bosch BME280, an integrated environmental sensor combining temperature, barometric pressure, and humidity measurement in a single compact package. This precision sensor is commonly used for context awareness, fitness monitoring, home automation control, weather forecasts, IOT and indoor and outdoor navigation. Some main specifications for each of its sensors include:

Humidity: $\pm 3\%$ accuracy

Barometric Pressure: ± 1 hPa absolute accuracy

Temperature: $\pm 1.0^{\circ}\text{C}$ accuracy

Altitude Resolution: ± 1 meter accuracy (when properly calibrated)

Communication Interfaces: The BME280 supports both I²C and SPI protocols, providing flexibility for integration with various microcontroller platforms.

Altitude Measurement: Because atmospheric pressure decreases predictably with altitude, the BME280's precise pressure measurements enable altimeter functionality. However, accurate altitude calculation requires knowing the current sea level pressure for the specific location and day. Without up-to-date calibration data, altitude accuracy typically degrades to approximately ± 10 meters.

The BME280's combination of small size, low cost, and multi-parameter sensing makes it ideal for projects ranging from wearables to portable weather stations to IoT environmental monitoring systems.

Datasheet and Considerations

Choosing the appropriate weather sensor requires evaluating several key factors, such as what you intend to measure, the operating environment, and the overall system that will interface with the sensor.

Operating environment: Consider where the sensor will operate, which may put constraints on other requirements, such as size or durability. For example, the sensor may be integrated into a wearable device where space is limited or mounted outdoors where weatherproofing is critical. A sensor with excellent measurement range and advanced features may not suit your installation constraints.

Measurement speed: Determine whether periodic sampling or continuous real-time monitoring is required. Some applications need instantaneous data (such as drone stabilization), where faster sampling rates may increase power consumption, while others can function with measurements taken every few minutes (such as climate logging).

Accuracy requirements: Data quality depends on measurement accuracy. For a personal weather station, the difference between $\pm 0.1^{\circ}\text{C}$ and $\pm 0.01^{\circ}\text{C}$ accuracy may be negligible. However, for business-critical applications like assessing workplace safety or developing weather preparedness plans, higher precision becomes essential.

Build quality and durability: Sensors must withstand their operating environment. If sensors cannot endure environmental wear or maintain consistent performance throughout their service life, they compromise data quality and system safety. For example, in high-strain applications—such as airport runway wind sensors or submerged flow meters—robustness and lifespan directly impact data reliability. Outdoor installations require weatherproof housings, UV resistance, and appropriate IP ratings.

Power consumption: Battery-powered or remote installations require low-power sensors with efficient sleep modes. Evaluate both active measurement current and standby power to estimate battery life for your deployment scenario.

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